Algebra II with Trigonometry Chapter 11.3

Mental Math (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file: 4666324_alg-2trig-h-chp-11-3-note-docx.pdf

Geometric Definition If the vertices of a hyperbola are located at (4, 0) and (-4, 0), what is the constant difference of the distances from any point on the hyperbola to its two foci?

Transverse Axis Length What is the length of the transverse axis for the hyperbola described by the standard equation $x^2/25 - y^2/16 = 1$?

Direction of Opening Does the hyperbola given by the equation $y^2/9 - x^2/4 = 1$ have a horizontal or vertical transverse axis?

Identifying Vertices What are the coordinate points of the vertices for the hyperbola with the equation $x^2/49 - y^2/100 = 1$?

Asymptote Slopes What are the slopes of the asymptotes for the hyperbola $x^2/9 - y^2/16 = 1$?

Calculating Focal Distance For a hyperbola centered at the origin with $a^2 = 9$ and $b^2 = 16$, what is the distance from the center to either of its foci?

Equation Building from Asymptotes If a hyperbola centered at the origin has vertices at (1, 0) and (-1, 0) and its asymptotes are $y = 5x$ and $y = -5x$, what is the value of b^2 in its standard equation?

Sketching the Central Box When drawing the central box to sketch the hyperbola $y^2/36 - x^2/4 = 1$, what are the total width (along the x-axis) and total height (along the y-axis) of the rectangle?

Locating the Foci On which axis (x or y) do the foci of the hyperbola $y^2/64 - x^2/81 = 1$ lie?

Pythagorean Relationship If a hyperbola has vertices at $(\pm 2, 0)$ and foci at $(\pm 6, 0)$, what is the value of $a^2 + b^2$ (also known as c^2) for this hyperbola?